MARMARA UNIVERSITY

FACULTY OF ENGINEERING

MECHANICAL ENGINEERING DEPARTMENT

ME 3061 FLUID MECHANICS

2022-2023 FALL TERM PROJECT

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Date: 08.01.2024

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1. Introduction

Purpose of this project is to design and optimize a hot water delivery system for a five-storey building with certain constraints and requirements. The system consists of a boiler, a pump, pipes of different diameters, valves, and radiators. The system should deliver water at 50°C to each floor at a flow rate of 0.3 m^3/h, while maintaining the flow velocity in the pipes between 0.5 m/s and 1 m/s. The system should also minimize the head loss, the power consumption, and the cost. To achieve this specifications, we need to select a suitable pump, pipe diameters, pipe material, type of valves, and operating conditions. We also need to perform a cost analysis for the system, excluding the boiler and the radiators.

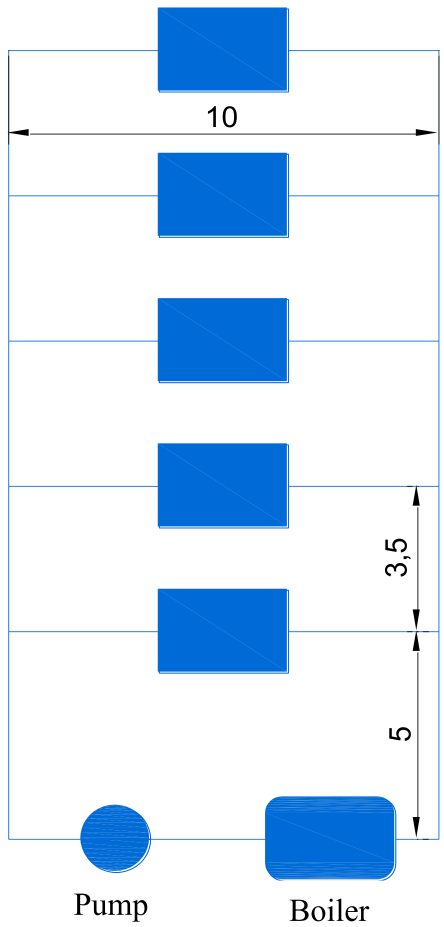

Figure 1 (Schematic of the hot water delivery system)

2. Analysis

The flow rate entering each floor, and the velocity range of the fluid is given. The diameter of the pipes, the head loss, and the pumping power requirements are to be determined.

Assumptions:

1.The flow is steady and incompressible. 2. The fluid temperature does not change thus flow is isothermal, so μ is constant. 3. The entrance effects are negligible, and thus the flow is fully developed. 4. The project instruction states that the minor losses due to cross-sectional area change is negligible.

Properties:

The density and the dynamic viscosity of the water at 50°C are given to be ρ= 988 kg/m^3 and μ=0.000546 kg/m-s, respectively.

Analysis:

The flow rate in each floor is given as 0.3 m^3/h. For a pipe that branches into parallel pipes and then rejoins at a junction downstream, the total flow rate is the sum of the flow rates in individual pipes.

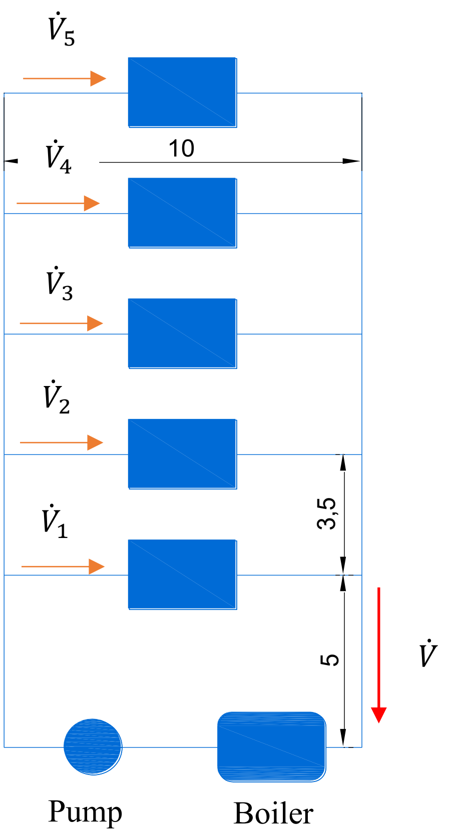

Figure 2 (Schematic of the volume flow rate distribution.-1)

Thus, the flow rate at a junction downstream is:

$$\dot{V} = \dot{V}_1 + \dot{V}_2 + \dot{V}_3 + \dot{V}_4 + \dot{V}_5 = (0.3+0.3+0.3+0.3+0.3)\ m^3/h = 1.5\ m^3/h \tag{1}$$

As the $\dot{V}$ goes upper floors the flow will split at each floor, and $\dot{V}$ will decrease by 0.3 m^3/h at each floor until it reaches to the top.

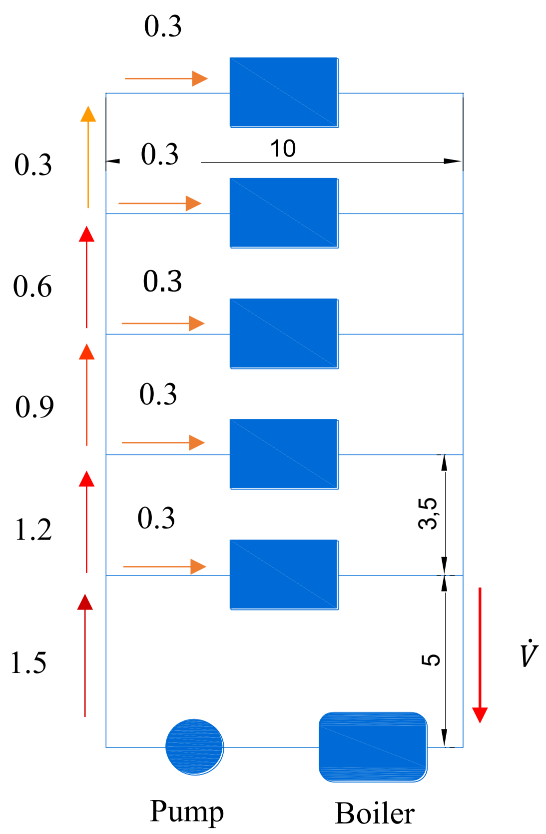

Figure 3 (Schematic of the volume flow rate distribution.-2)

As we determined the flow rate in each pipe, then we need to select appropriate pipe diameter which is suitable with the given range of fluid velocity and the determined flow rates.

$$V = \frac{\dot{V}}{A_c} = 0.5 - 1\ m/s \tag{2}$$

The each diameters from Fırat Boru catalogue were bring into equation for each flow rate value. Then the diameters were determined as in the table:

FIRATTHERM PN25 PPRC BORULAR				
KODU	ÇAP	S:Et Kalınlığı	AMBALAJ	FİYAT TL /METRE
7B00020020	20	3.4	100 Metre / Bağ	13,09
7B00020025	25	4.2	80 Metre / Bağ	21,57
7B00020032	32	5.4	40 Metre / Bağ	40,91
7B00020040	40	6.7	40 Metre / Bağ	58,32
7B00020050	50	8.4	20 Metre / Bağ	93,11
7B00020063	63	10.5	20 Metre / Bağ	149,29
7B00020075	75	12.5	12 Metre / Bağ	225,48
7B00020090	90	15.0	8 Metre / Bağ	359,65
7B00020110	110	18.3	4 Metre / Bağ	535,32
7B00020126	125	20.8	4 Metre / Bağ	684,62
7B00020127	160	26.6	4 Metre / Bağ	1.190,87

Figure 4 (PN25 PPRC Pipe Informations.)

No\Property	Diameter (m)
1	0,0266
2	0,0266
3	0,0132
4	0,0266
5	0,0132
6	0,0212
7	0,0132
8	0,0166
9	0,0132
10	0,0132
11	0,0132

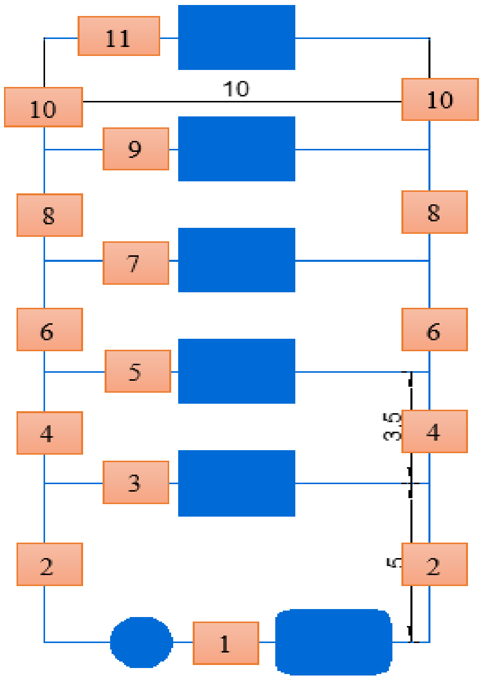

Figure 5 (Pipe diameters of the system.)

After that the velocities were calculated using equation (2) for all pipes. And the Reynolds numbers were calculated using the velocity values. The velocity and Reynolds number of the fluid in the pipe no.1 is calculated as following:

$$V_1 = \frac{\dot{V}_1}{A_{c_1}} = \frac{0.0004167\ m^3/s}{0.000556\ m^2} = 0.75\ m/s \qquad (3)$$

$$Re_1 = \frac{\rho V_1 D_1}{\mu} = \frac{(988\frac{kg}{m^3})(0.75\ m/s)(0.0266\ m^2)}{0.000546\ \mathrm{kg/(m.s)}} \qquad (4)$$

All velocities and Reynolds numbers were calculated using excel and the values are shown in the table:

No\Property	Diameter (m)	Flow Rate (m^3/s)	Velocity (m/s)	Re
1	0,0266	0,0004167	0,750	36024,31
2	0,0266	0,0004167	0,750	36024,31
3	0,0132	0,0000833	0,609	14518,89
4	0,0266	0,0003333	0,600	28819,45
5	0,0132	0,0000833	0,609	14518,89
6	0,0212	0,0002500	0,708	27120,19
7	0,0132	0,0000833	0,609	14518,89
8	0,0166	0,0001667	0,770	23090,28
9	0,0132	0,0000833	0,609	14518,89
10	0,0132	0,0000833	0,609	14518,89
11	0,0132	0,0000833	0,609	14518,89

Figure 6 (Flow rate, Velocity and Reynold's Number calculations.)

After that the Darcy friction factor (f) were calculated using Haaland's Equation:

$$\frac{1}{\sqrt{f}} = -1.8 \log\left(\frac{6.9}{Re} + \left(\frac{\varepsilon/D}{3.7}\right)^{1.11}\right) \tag{5}$$

Plastic FIRATTHERM PPRC pipes were chosen for the system. Thus, the relative roughness for the plastic the pipe (ε) can be neglected. All Darcy friction factor values calculated in excel and made a table as follows:

No\Property	f
1	0,0223
2	0,0223
3	0,0279
4	0,0235
5	0,0279
6	0,0239
7	0,0279
8	0,0248
9	0,0279
10	0,0279
11	0,0279

Figure 7 (Darcy Friction Factor calculations.)

All major head losses were computed for all pipe sections using following equation in excel:

$$h_{L,major} = f\frac{L}{D}\cdot\frac{V^2}{2g} \tag{6}$$

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=(1/(-1,8*LOG10(((0/3,7)^1,1)+(6,9/E78))))^2
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Pipes	Length (m)	h_loss,major (m)
1	10	0,240539275
2	5	0,120269637
3	3,5	0,140064932
4	3,5	0,100828113
5	3,5	0,158338512
6	3,5	0,140064932
7	10	0,400185519

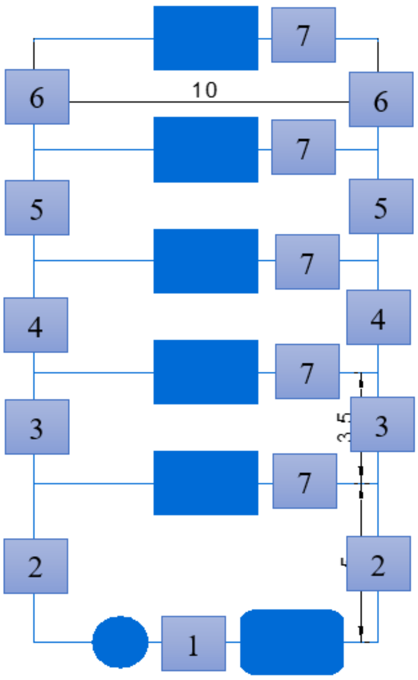

Figure 8 (Major head losses.)

Then all minor head losses in the system were calculated using following equation in excel. The loss coefficient for bends taken as, 0.3.

$$h_{L,minor} = K_L \frac{V^2}{2g} \tag{7}$$

Bends	h_loss,minor (m)
1	0,00860
2	0,00567

Figure 9 (Minor head losses in bends.)

The loss coefficient for tees in vertical direction taken as, 0.2, and the loss coefficient in horizontal direction taken as 1. The minor head losses were calculated using equation (7) in excel.

Tees	h_loss,Vertical (m)	h_loss,Horizontal (m)
1	0,0057	0,0287
2	0,0037	0,0183
3	0,0051	0,0256
4	0,0060	0,0302

Figure 10 (Minor head losses in tees.)

The loss coefficient for radiator is taken as 7 as stated in the specifications. The minor head losses of radiators were calculated using equation (7) in excel.

Radiators	h_loss,minor (m)
All	0,132

Figure 11 (Minor head loss in radiator.)

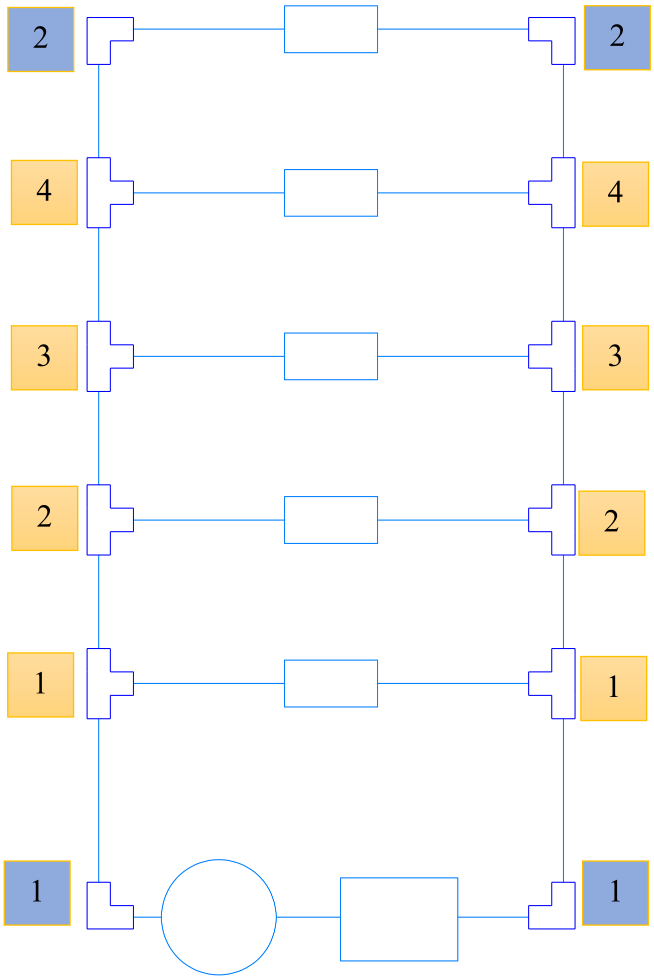

Figure 12 (Names of tees and bends.)

The pressure drop (or head loss) in each individual pipe connected in parallel must be same since $\Delta P = P_A - P_B$ and junction pressures P_A and P_B are same for all the individual pipes. In our system the flow rates at each floor is the same therefore the head losses must be the same in each floor. In order to do that we need to put valves to increase the head loss when needed.

The system is analyzed junction by junction starting from the top floor. Then the head losses between two floor made equal by putting valve.

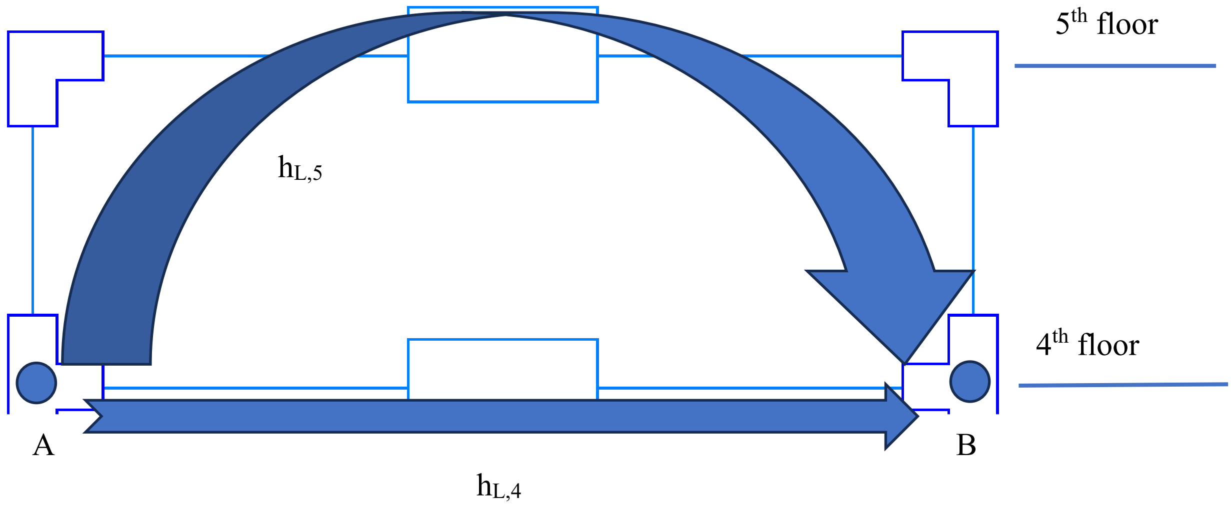

Figure 13 (Junction analysis.)

$$h_{L,5} = h_{L,4} \tag{8}$$

The head loss values calculated in previous section were used to evaluate the equations the difference between two head loss values between junctions is the head loss value of the valve. To find the head loss of the valve in each floor and same procedure were used between other junctions in the system at downwards direction. The computation were made in excel.

Loop	h_loss,branch1	h_loss,branch2	h_loss,valve	k_valve
1 (5-4)	0,836	0,593	0,243	12,863
2 (4-3)	1,163	0,584	0,579	30,652
3 (3-2)	1,372	0,569	0,803	42,475
4 (2-1)	1,664	0,590	1,074	56,811

Figure 14 (Total head losses and required valve constants calculation in each loop.)

Finally, to find the required pumping power we need to write the energy equation between point A and B.

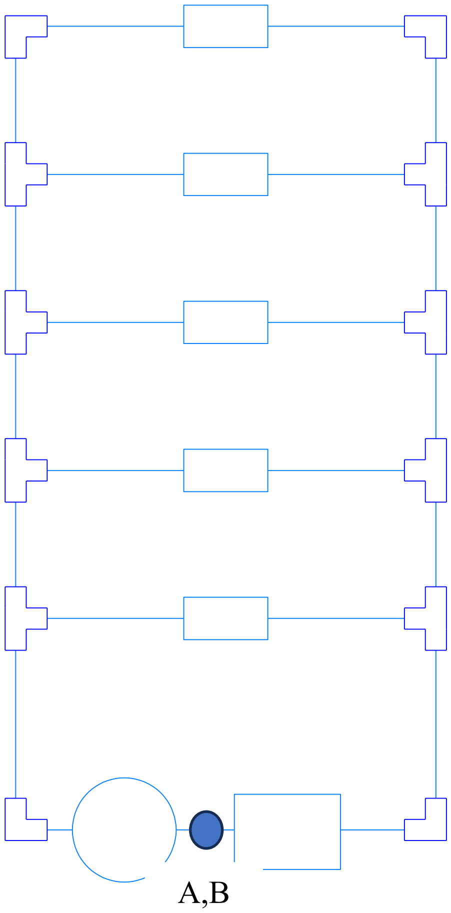

Figure 15 (Energy equation analysis points.)

$$\frac{P_A}{\gamma_A} + \frac{V_A^2}{2g} + z_A + h_P = \frac{P_B}{\gamma_B} + \frac{V_B^2}{2g} + z_B + h_L \tag{9}$$

Point A=B. Thus,

$z_A = z_B$ $\quad$ $P_a = P_B$ $\quad$ $V_A = V_B$

When we rearrange the energy equation:

$$h_P = h_L \tag{10}$$

We compute the total head loss and equate it to pump power using excel.

h_loss,total = h_pump		Unit
h_loss,total	7,196271889	m

Figure 16 (Total head loss.)

To convert the unit into watts we can use this equation:

$$P = h_P \cdot \gamma \cdot \dot{V} = (7.20\text{ m})(9692.48)(0.000417\text{m}^3/\text{s}) = 29.06\text{ Watt} \quad (11)$$

To obtain this power Mars MRS 25/6-130 were chosen

Model Adı	Genişlik (B)	Uzunluk (H)	Derinlik (L)	Çap (D)	Ağırlık	Voltaj (V)	Güç (W)	Akım (A)	Basma Yüksekliği	Debi (m^3/ h)
MRS 25/4 - 130	130 mm	130 mm	126 mm	11/2"	2,4 kg	220 V	22	0,17	4 m	2,5
MRS 25/6 - 130 **MRS 25/6 - 180**	130 mm	130 mm 180 mm	126 mm	11/2"	2,4 kg	220 V	45	0,35	6 m	2,5

Figure 17 (Selected pump and specifications.)

3. Cost Analysis

Component	Quantity	Unit Cost	Total Cost
FIRATTHERM PPRC Pipe 20 mm Dia	57 m	13.09 TL/m	746.13 TL
FIRATTHERM PPRC Pipe 25 mm Dia	14 m	21,57 TL/m	301.98 TL
FIRATTHERM PPRC Pipe 32 mm Dia	7 m	40,91 TL/m	286.37 TL
FIRATTHERM PPRC Pipe 40 mm Dia	27 m	58,31 TL/m	1574.37 TL
Bend 40mm Dia	2	9,92 TL	19.84 TL
Bend 20mm Dia	2	1.63 TL	3.26 TL
Tee	6	4.12 TL	24.72 TL
Global Valve	4	51.50 TL	206 TL
Mars MRS 25/6-130 Pump	1	3529.16 TL	3529.16 TL
		Overall Total Cost:	6691.83 TL

Table 1 (Cost Analysis Table)

4. Conclusion

In this project, we designed and optimized a hot water delivery system for a five-storey building. We considered the following criteria: water temperature, flow rate, flow velocity, head loss, power consumption, and cost. We selected the optimal pump, pipe diameters, pipe material, type of valves, and operating conditions that met the requirements and constraints. We also performed a cost analysis for the system, excluding the boiler and the radiators.

The results showed that our system can deliver water at 50°C to each floor at a flow rate of 0.3 m3/h, while maintaining the flow velocity in the pipes between 0.5 m/s and 1 m/s. The system has a low head loss, a low power consumption, and a reasonable cost. The system is also reliable, safe, and easy to maintain.

The project demonstrated the application of engineering principles and methods to design and optimize a hot water delivery system. The project also enhanced our skills in problem-solving, modeling, analysis, and decision-making. The project was a valuable learning experience that prepared us for future engineering challenges.

5. References

[1] Fluid mechanics : fundamentals and applications / Yunus A. Çengel, John M. Cimbala.—1st ed. p. cm.—(McGraw-Hill series in mechanical engineering)
[2] Fırat Boru Fıratherm PPRC Kompozit Boru (PN20-PN25)